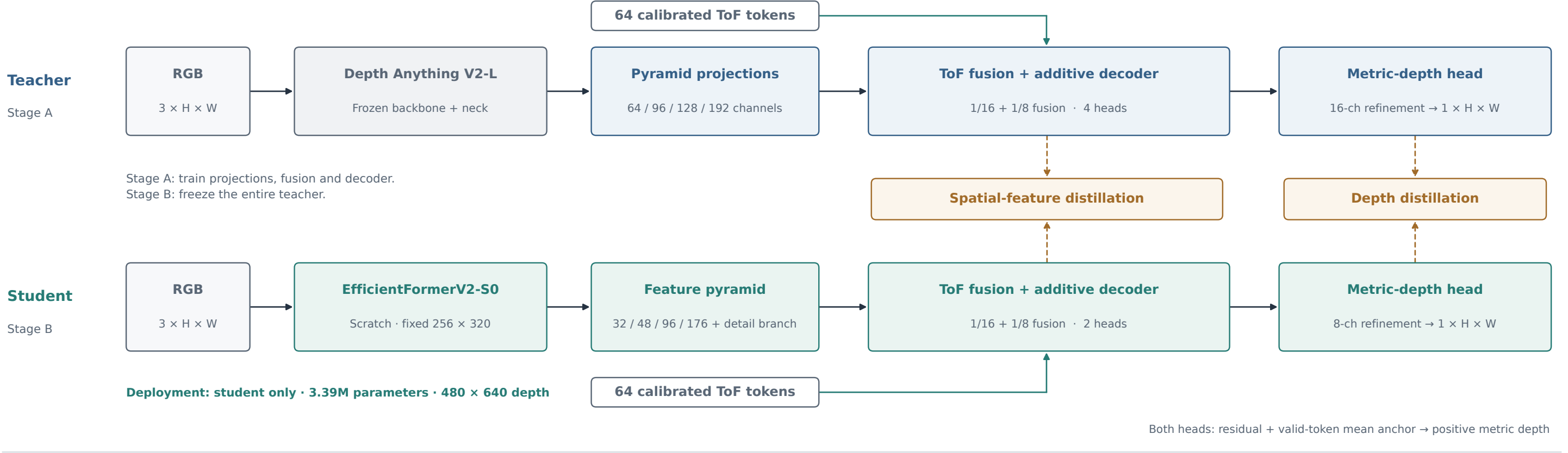
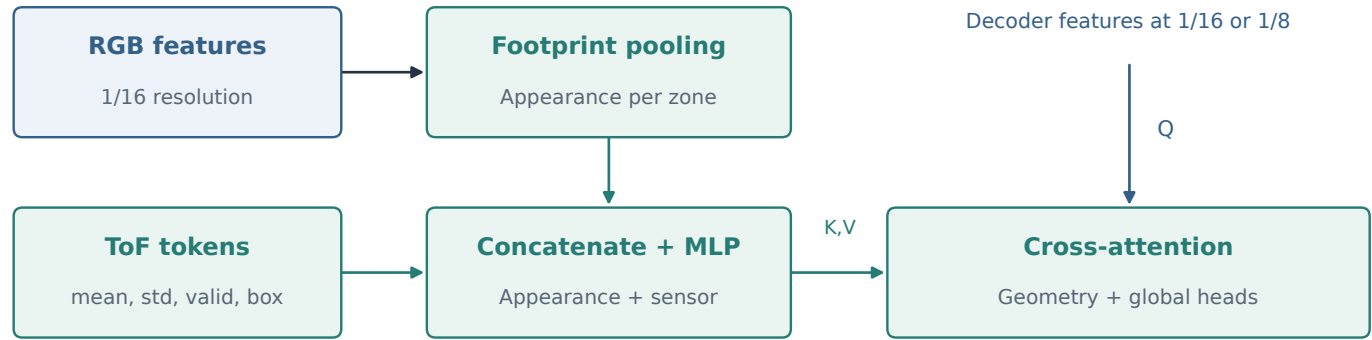


(a) RGB-ToF teacher-student framework

Input / output resolution: 640 × 480



(b) Appearance-enriched sensor fusion



Separate teacher / student weights · invalid-token mask · learnable null token

→ Feature flow      - - - - - Training supervision

(c) Coverage-aware knowledge transfer

Teacher targets are detached; gradients update the student and adapters.

$$\mathcal{L} = \mathcal{L}_{\text{MSE}} + 0.1\mathcal{L}_{\text{grad}} + r(e)[0.5\mathcal{L}_{\text{depth}} + 0.05\mathcal{L}_{\text{feat}}]$$

Depth: Smooth L1 · Features: cosine loss after 1 × 1 adapters

Coverage weights: outside 2 / inside 1; area-pooled for feature transfer.

Confidence:  $\exp(-|D_T - D_{GT}|/0.25)$  attenuates the teacher gradient.

**Schedule:** GT only (1-5) → ramp (6-10) → full distillation

Fusion is interleaved with decoding; RGB skip connections are summarized.